Dual-ligand screen / Setup

48 pairs, then 22 pairs + L17 reference. One 48 × 2 mL Para-Dox, 71 fresh reaction vials. Manual caps and labware moves. **All precise liquid handling is robotic**, including stocks, standards, each round's dosing and ligand refills.

Load chemicals

- Weigh labeled charges from `weighed_charges_template.csv`; enter actual masses/assays. Weigh ligands in the glovebox. Robot adjusts solvent addition; respect mass limits.
- Coarse-load raw solvents to CSV minimum fills. Supplied **8 ×30 mL A1-H1 reservoir: single-channel ONLY**. No manual aliquoting or volumetric top-up.
- **Release blocked:** default bulk sources fail the conservative no-nozzle-entry check. Supply actual short, solvent-compatible stock containers/rack JSON; regenerate and check reach. Carrier slots **3/6**; retain capped reserves and original working tubes.
- Have **three compatible 96-well LC plates**, 2.2 mL geometry, 1 mL/sample, carrier/seals. Confirm offsets, mounting, caps and actual 1.7-min method/cycle. Import/simulate locally.

Pipettes

Initial preparation: p300 single left /p1000 single right; 300 µL tips in 5, 1000 µL in 8/9. After preparation, change RIGHT to **p300 eight-channel GEN2** and tips 8/9 to 300 µL via the App installation/calibration workflow. Left stays installed.

Stocks - all liquids prepared by robot

Stock	Rule
Ligands	Nominal **20 mM**, generous reserves. Weighed charges in mapped carriers.
RockPhos L03	**~7.50 mg pure equivalent**, assay-corrected; 10 mg available. Robot makes **~800 µL in 1 C1**. Retain SAME tube; **max 4 R2 wells**.
Pd(COD)(DQ)	Direct in **4 A3/B3/C3: 22.8 mg each; A4: 21.6 mg**. Robot adds ~1.50/~1.43 mL toluene, **40 mM**. **90 mg total; 10 mg for scope.** Retain all four; no Pd refill.
SM + mesitylene	**~322.32 mg SM** at 100% assay; robot adds neat IS/toluene: nominal **1 M /100 mM**, ~2 mL. **4 D2: 1.89 mL** for both rounds; retain capped between rounds, **no SM refill**.

Concentrations use nominal solvent addition. **No hand refills or R2 stock remake.**

Reaction-stage deck

Slot	Contents
1 /4	2 mL working tubes, max 1.95 mL. Neat IS **1 C2**, SM/IS **4 D2**, Pd **4 A3/B3/C3/A4**; prep acid **4 D3**.
2	EMPTY 12-channel reservoir; robot makes quench A4/A6/A9 and diluent A5/A7/A10/A11/A12.
3 /10	Reaction block: pipetting /heater-shaker. During holds, raw solvents temporarily occupy 3.
4 /6 in R1 hold	Prepared analytical rack temporarily in 4; calibration plate in 6, then directly to LC. Later, 6 takes each final LC plate.
5 /8 /9	Tips. Keep **7 and 11 empty**.

Dual-ligand screen / Run

Operator / start: _____ LC method / channel: _____

Use regenerated files for the current R1 design; discard earlier dosing exports. Follow App pauses and matching loading CSVs. Staff handle weighing, coarse loading, caps, moves and local starts. Preparation counts inside the campaign.

Step	Action
1 / Stocks	Run `prepare_stocks.py` from actual weights. Load requested carriers uncapped; cap/store when removed. Retain the preparation manifest. Make the one right-pipette/tip-rack change after preparation.
2 / R1 setup	Load **48 NEW EMPTY vials** in the reaction block at 3. Uncap the required stock tubes; start R1 Python.
3 / R1 reaction	Robot doses **25 µL SM/IS +75 µL ligand A +75 µL ligand B +75 µL Pd**. Hand-cap; move/latch. **95 °C, 500 rpm, 4 h. Start the external timer when the App announces the hold starts.**
4 / During R1 hold	Leave reaction on 10. Load raw solvents in 3 and analytical rack in 4 at prompts. Robot makes **14 standards** and quench/diluent in 2. Seal/move calibration plate **6** → **HPLC**; start sequence. Restore rack 4; remove raw reservoir. **Monitor the timer:** pauses do not stop heating. If the deadline arrives while paused, stop heat/shaking via App and record the overrun; do not resume dosing.
5 / Cool / quench	Sealed block to 3; cool. **Keep capped while robot pre-fills LC wells with 980 µL diluent.** Then uncap; robot adds **500 µL quench**. Recap ALL; shake **1500 rpm, 2 min**, heater off; return to 3.
6 / LC samples	Let bubbles clear; uncap indicated vials. Robot promptly adds **20 µL extract to prefilled LC wells**, fresh sample tips. Recap reaction vials.
7 / Final plate	Move **6** → **cooled shaker 10**. Robot shakes **UNCOVERED, heater OFF, 1000 rpm, 2 min** at confirmed splash-free speed. Move **10** → **HPLC**, seal and inject directly. No manual dilution/vial transfers.
8 / R2 select/refill	After all **48 valid R1 results**, Claude selects **22 new pairs +L17 reference**. Run `refill_stocks.py` **FIRST** for ligand top-ups from original carriers. Retain RockPhos, SM/IS **4 D2** and all FOUR Pd tubes; **no SM/Pd refill**.
9 / R2	Load **23 NEW EMPTY vials** in the SAME block; replace reservoir 2 with a NEW EMPTY one. Uncap required stocks; run R2 Python: **25 µL SM/IS each, then catalysts**. Reference uses **150 µL L17**, no B. Hand-cap; repeat reaction/workup, with fresh robot-made workup fluids during the hold. **R2 H3 is diluent only: do not inject.**

Calibration: each analyte blank +**10, 50, 150, 350, 800 µM**, independent **666.7 µM** check; constant **66.7 µM mesitylene**. Python calculates calibrated yields from integrated UV areas. Stop the clock after all 23 final injections and quantitative analysis.

Pause for blocked tips, persistent bubbles, air uptake, obvious layers or tip contact. Do not guess liquid top-ups or repair a dilution by hand. Report deviations so stock accounting can be corrected.

Timing: 12 h is advisory, not demonstrated. Record actual times and hold deviations. Give logs/raw LC files/stock records to the pipeline operator; return using `python scripts/hte-return-package.py`.